



## Week 2 Status Report

**Project:** CCP-CN-S2026 – Problem 10: High Availability Router

**Team Name:** Guardian Gateways

**Week2:** (GNS3 Implementation & Basic Connectivity)

**Date:** 20, May 2026

| MEMBERS NAME        | STUDENT ID  |
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## Scenario:

A water treatment plant cannot afford downtime. Two routers must provide gateway redundancy: if Router A fails, Router B takes over automatically within a few seconds.

## Devices Used

- 2 Cisco 4331 Routers
- 1 Core Switch
- 2 VPCS

## 1. Tasks Completed This Week

- ❖ Full network topology successfully built in GNS3.
- ❖ IP addressing completed on all devices.
- ❖ HSRP configured with priority, preempt, and interface tracking.
- ❖ Basic router security and SSH access implemented.
- ❖ All-to-All connectivity tested successfully.
- ❖ Running-configs of both routers are saved and uploaded to GitHub.
- ❖ Python environment setup completed.

## 2. Major Decisions Taken

- ❖ Used HSRP for gateway redundancy.
- ❖ LAN Subnet: 192.168.10.0/24
- ❖ Virtual Gateway IP: 192.168.10.1
- ❖ R1 as Active Router (Priority 120), R2 as Standby Router (Priority 100).

## 3. Challenges Faced

- ❖ IOS image compatibility issues in GNS3.
- ❖ Initial HSRP failover testing required troubleshooting.
- ❖ Limited team size (5 members) increased individual workload.
- ❖ IP addressing mismatch between Kali Linux and router network.

## 4. Deliverables Completed

| Deliverable       | Status    | Remarks                 |
|-------------------|-----------|-------------------------|
| GNS3 Project File | Completed | Full topology with HSRP |

|                           |           |                           |
|---------------------------|-----------|---------------------------|
| Running-configs (R1 & R2) | Completed | Uploaded on GitHub        |
| Ping Test Results         | Completed | All-to-All matrix created |
| Screenshots               | Completed | See below                 |
| Week 2 Status Report      | Completed | This document             |

## 5. Screenshots

Figure 1: Complete GNS3 Topology

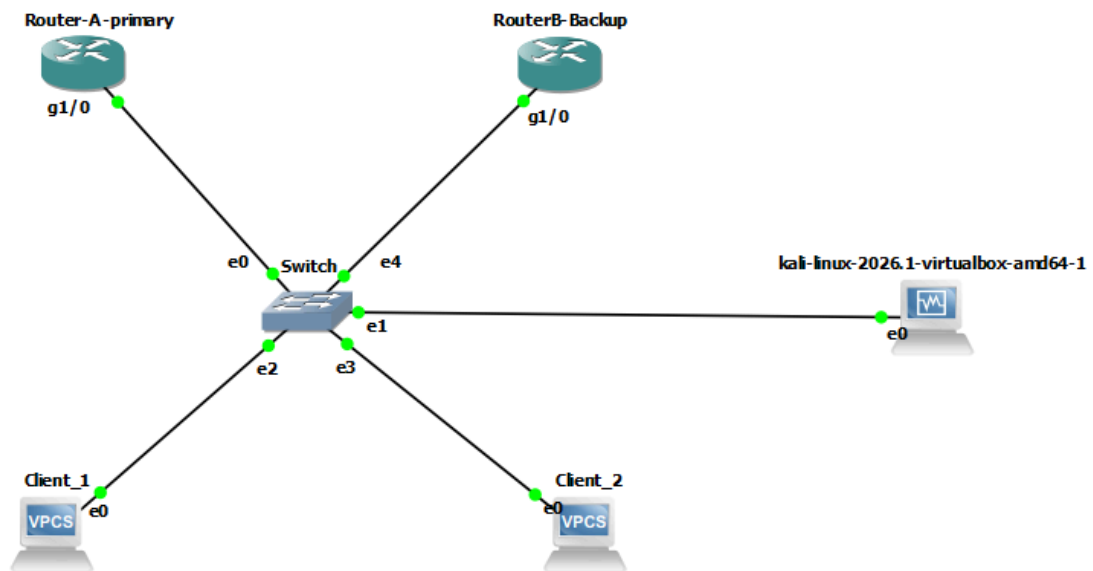


Figure 2: R1 show ip interface Brief

```
Jun 13 00:34:34.331: %SYS-5-CONFIG-I: Configured from console by console
R1#show ip interface brief
Interface                IP-Address      OK? Method Status          Protocol
FastEthernet0/0          unassigned      YES NVRAM    administratively down down
GigabitEthernet1/0       192.168.10.2    YES manual    up              up
R1#config
```

Figure 3: R2 show ip interface Brief

```
# type show for a list of subcommands
R2#show ip interface brief
Interface                IP-Address      OK? Method Status          Protocol
FastEthernet0/0          unassigned      YES NVRAM    administratively down down
GigabitEthernet1/0       192.168.10.3    YES NVRAM    up              up
R2#
*Jun 13 00:41:55.215: %HSRP-5-STATECHANGE: GigabitEthernet1/0 Grp 1 state Active -> Spea
```

**Figure 4: R1 show standby brief**

```
% Type "show ?" for a list of subcommands
R1#show standby brief
                P indicates configured to preempt.
                |
Interface   Grp  Pri P State   Active        Standby        Virtual IP
Gi1/0      1    120 P Active  local         192.168.10.3   192.168.10.1
R1#con
```

**Figure 5: R2 show standby brief**

```
R2#show standby brief
                P indicates configured to preempt.
                |
Interface   Grp  Pri P State   Active        Standby        Virtual IP
Gi1/0      1    100 P Standby 192.168.10.2   local         192.168.10.1
R2#
```

**Figure 6: R1 show track**

```
R1#show track
Track 1
  Interface GigabitEthernet1/0 line-protocol
  Line protocol is Up
    1 change, last change 00:03:00
  Tracked by:
    HSRP GigabitEthernet1/0 1
R1#
```

**Figure 7: R1 show track**

```
R2#show track
Track 1
  Interface GigabitEthernet1/0 line-protocol
  Line protocol is Up
    1 change, last change 00:01:19
  Tracked by:
    HSRP GigabitEthernet1/0 1
R2#
```

**Figure 8: Client1 Ping to Virtual IP (192.168.10.1)**

```
VPCS> ping 192.168.10.1
84 bytes from 192.168.10.1 icmp_seq=1 ttl=255 time=31.190 ms
84 bytes from 192.168.10.1 icmp_seq=2 ttl=255 time=16.616 ms
84 bytes from 192.168.10.1 icmp_seq=3 ttl=255 time=16.306 ms
84 bytes from 192.168.10.1 icmp_seq=4 ttl=255 time=15.626 ms
84 bytes from 192.168.10.1 icmp_seq=5 ttl=255 time=40.954 ms
```

**Figure 9: Client1 Ping to Client2 (192.168.10.20)**

```
VPCS> ping 192.168.10.20
84 bytes from 192.168.10.20 icmp_seq=1 ttl=64 time=1.061 ms
84 bytes from 192.168.10.20 icmp_seq=2 ttl=64 time=1.014 ms
84 bytes from 192.168.10.20 icmp_seq=3 ttl=64 time=1.325 ms
84 bytes from 192.168.10.20 icmp_seq=4 ttl=64 time=0.790 ms
84 bytes from 192.168.10.20 icmp_seq=5 ttl=64 time=1.190 ms
```

**Figure 10: Client2 Ping to Client1(192.168.10.10)**

```
VPCS> ping 192.168.10.10
84 bytes from 192.168.10.10 icmp_seq=1 ttl=64 time=0.838 ms
84 bytes from 192.168.10.10 icmp_seq=2 ttl=64 time=0.686 ms
84 bytes from 192.168.10.10 icmp_seq=3 ttl=64 time=1.556 ms
84 bytes from 192.168.10.10 icmp_seq=4 ttl=64 time=0.728 ms
84 bytes from 192.168.10.10 icmp_seq=5 ttl=64 time=0.583 ms
```

**Figure 11: Router1 Active and Router2 Standby**

```
R1#show standby brief
          P indicates configured to preempt.
          |
Interface  Grp  Pri P State  Active        Standby        Virtual IP
Gi1/0      1    120 P Active  local         192.168.10.3   192.168.10.1
R1#
```

```
R2#show standby brief
          P indicates configured to preempt.
          |
Interface  Grp  Pri P State  Active        Standby        Virtual IP
Gi1/0      1    100 P Standby 192.168.10.2   local          192.168.10.1
R2#
```

**Figure 11: Continuous ping running**

```

VPCS> ping 192.168.10.20 -t
84 bytes from 192.168.10.20 icmp_seq=1 ttl=64 time=7.110 ms
84 bytes from 192.168.10.20 icmp_seq=2 ttl=64 time=1.199 ms
84 bytes from 192.168.10.20 icmp_seq=3 ttl=64 time=0.621 ms
84 bytes from 192.168.10.20 icmp_seq=4 ttl=64 time=0.673 ms
84 bytes from 192.168.10.20 icmp_seq=5 ttl=64 time=0.667 ms
84 bytes from 192.168.10.20 icmp_seq=6 ttl=64 time=0.922 ms
84 bytes from 192.168.10.20 icmp_seq=7 ttl=64 time=0.731 ms
84 bytes from 192.168.10.20 icmp_seq=8 ttl=64 time=0.886 ms
84 bytes from 192.168.10.20 icmp_seq=9 ttl=64 time=0.707 ms
84 bytes from 192.168.10.20 icmp_seq=10 ttl=64 time=0.950 ms
84 bytes from 192.168.10.20 icmp_seq=11 ttl=64 time=0.898 ms
84 bytes from 192.168.10.20 icmp_seq=12 ttl=64 time=0.680 ms
84 bytes from 192.168.10.20 icmp_seq=13 ttl=64 time=0.882 ms
84 bytes from 192.168.10.20 icmp_seq=14 ttl=64 time=0.669 ms
84 bytes from 192.168.10.20 icmp_seq=15 ttl=64 time=0.697 ms
84 bytes from 192.168.10.20 icmp_seq=16 ttl=64 time=0.683 ms
84 bytes from 192.168.10.20 icmp_seq=17 ttl=64 time=0.701 ms
84 bytes from 192.168.10.20 icmp_seq=18 ttl=64 time=0.674 ms

84 bytes from 192.168.10.20 icmp_seq=19 ttl=64 time=0.716 ms
84 bytes from 192.168.10.20 icmp_seq=20 ttl=64 time=0.657 ms
84 bytes from 192.168.10.20 icmp_seq=21 ttl=64 time=0.657 ms
84 bytes from 192.168.10.20 icmp_seq=22 ttl=64 time=0.686 ms
84 bytes from 192.168.10.20 icmp_seq=23 ttl=64 time=0.675 ms
84 bytes from 192.168.10.20 icmp_seq=24 ttl=64 time=0.780 ms
84 bytes from 192.168.10.20 icmp_seq=25 ttl=64 time=0.703 ms
84 bytes from 192.168.10.20 icmp_seq=26 ttl=64 time=0.694 ms
84 bytes from 192.168.10.20 icmp_seq=27 ttl=64 time=0.722 ms
84 bytes from 192.168.10.20 icmp_seq=28 ttl=64 time=0.754 ms
84 bytes from 192.168.10.20 icmp_seq=29 ttl=64 time=0.700 ms
84 bytes from 192.168.10.20 icmp_seq=30 ttl=64 time=0.646 ms
84 bytes from 192.168.10.20 icmp_seq=31 ttl=64 time=1.131 ms
84 bytes from 192.168.10.20 icmp_seq=32 ttl=64 time=0.673 ms
84 bytes from 192.168.10.20 icmp_seq=33 ttl=64 time=0.866 ms
84 bytes from 192.168.10.20 icmp_seq=34 ttl=64 time=0.685 ms
84 bytes from 192.168.10.20 icmp_seq=35 ttl=64 time=1.202 ms
84 bytes from 192.168.10.20 icmp_seq=36 ttl=64 time=0.690 ms
84 bytes from 192.168.10.20 icmp_seq=37 ttl=64 time=0.668 ms
84 bytes from 192.168.10.20 icmp_seq=38 ttl=64 time=0.661 ms
84 bytes from 192.168.10.20 icmp_seq=39 ttl=64 time=0.692 ms
84 bytes from 192.168.10.20 icmp_seq=40 ttl=64 time=0.739 ms
84 bytes from 192.168.10.20 icmp_seq=41 ttl=64 time=0.625 ms

```

**Figure 12: Continuous ping with very few packet losses in Kali.**

```

valid_ttl forever preferred_ttl forever
(kali㉿kali)-[~]
└─$ ping 192.168.10.2
PING 192.168.10.2 (192.168.10.2) 56(84) bytes of data.
64 bytes from 192.168.10.2: icmp_seq=1 ttl=255 time=957 ms
64 bytes from 192.168.10.2: icmp_seq=2 ttl=255 time=173 ms
64 bytes from 192.168.10.2: icmp_seq=3 ttl=255 time=52.4 ms
64 bytes from 192.168.10.2: icmp_seq=4 ttl=255 time=10.1 ms
64 bytes from 192.168.10.2: icmp_seq=5 ttl=255 time=9.12 ms
^C
— 192.168.10.2 ping statistics —
5 packets transmitted, 5 received, 0% packet loss, time 4004ms
rtt min/avg/max/mdev = 9.121/240.360/957.444/363.476 ms

```

**Figure 13: Router B becomes Active.**

```

R2#
R2#show standby brief
                P indicates configured to preempt.
                |
Interface    Grp  Pri  P State   Active        Standby        Virtual IP
Gi1/0        1    100 P Active local         unknown        192.168.10.1
R2#show s
*Jun 13 01:03:29.323: %HSRP-5-STATECHANGE: GigabitEthernet1/0 Grp 1 state Active -> Speakand
GigabitEthernet1/0 - Group 1

```

**Figure 14: Secure Router Access**

```

R1(config-if)#
R1(config-if)#enable secret cisco
R1(config)#line console 0
R1(config-line)#password cisco
R1(config-line)#login
R1(config-line)#service password-encryption
R1(config)#end
R1#show
*Jun 13 01:06:06.075: %SYS-5-CONFIG_I: Configured from console by consolerunning
Building configuration...

Current configuration : 951 bytes
!
! Last configuration change at 01:06:06 UTC Sat Jun 13 2026
!
version 15.2
service timestamps debug datetime msec
service timestamps log datetime msec
service password-encryption
!
hostname R1
!
boot-start-marker
boot-end-marker
!
!
enable secret 5 $1$y42f0DpKQv0p01C8ueC1Y0B/6H/

```

**Figure 15: Configure SSH**

```

% Type "show ?" for a list of subcommands
R1#show ip ssh
SSH Enabled - version 1.99
Authentication timeout: 120 secs; Authentication retries: 3
Minimum expected Diffie Hellman key size : 1024 bits
IOS Keys in SECSH format(ssh-rsa, base64 encoded):
ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAQgQD6BiI0CKPCVLQBFaBIvymhj4NFJUKmZDNNkz4M+M
qTV0hSUrMYyQG3WnYit0pBvt8P1if3dWFspZyC5A0/CcJWtwhGRmi2/B170Ft6ksQ9go3T/4opmPqX3K
fCFuGp9RZl1TrrtHR8ybdk50gfkYwP1FmM4bOpC2eQkzkH5w==
R1#

```

```

R2#show ip ssh
SSH Enabled - version 1.99
Authentication timeout: 120 secs; Authentication retries: 3
Minimum expected Diffie Hellman key size : 1024 bits
IOS Keys in SECSH format(ssh-rsa, base64 encoded):
ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAQgQCn3Wz9RjiYyqZB9qhoeJ6m9mle7Yk7sJXmsXItUzf4
AjiXrobWb8hBVFBNO8ETJXyjq1r3E4gNltyverrcHEkf5EEM+8JksUyS7LaFATkZLj0nkRyLETj0AA9
iIgrEw3fVFKM1Tu6oNVW03Ry2vCIKZWbLiAi2D8yCOeENq8Cew==

```

**Figure 16: SSH Login Successful from Client to Routers**

```
(kali㉿kali)-[~]  
$ ssh admin@192.168.10.2  
Unable to negotiate with 192.168.10.2 port 22: no matching key exchange method found. Their offer: diffie-hellman-group-exchange-sha1,diffie-hellman-group14-sha1,diffie-hellman-group1-sha1
```

Network connectivity between Kali Linux and RouterA was successfully verified through ICMP ping testing; however, SSH authentication verification could not be completed due to a client-side access issue.

**Show All-to-All Ping Matrix**

| Source \ Destination | R1<br>(192.168.10.2) | R2<br>(192.168.10.3) | Client1<br>(192.168.10.10) | Client2<br>(192.168.10.20) |
|----------------------|----------------------|----------------------|----------------------------|----------------------------|
| Client1              | Success              | Success              | –                          | Success                    |
| Client2              | Success              | Success              | Success                    | –                          |
| Router R1            | –                    | Success              | Success                    | Success                    |
| Router R2            | Success              | –                    | Success                    | Success                    |

**6. Next Week Plan (Week 3)**

- ❖ Implement advanced security features (ACLs, Port Security)
- ❖ Attack simulation (DoS attack)
- ❖ Mitigation strategies and testing
- ❖ PCAP capture and analysis

**Conclusion**

- High Availability achieved successfully